AI-Powered Career Growth and Productivity Platform

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INTRODUCTION

- Modern hiring uses automated systems (ATS) to screen and rank candidates, making resume quality and presentation critical.
- Many students struggle with ATS optimization, resume writing, and real interview preparation.

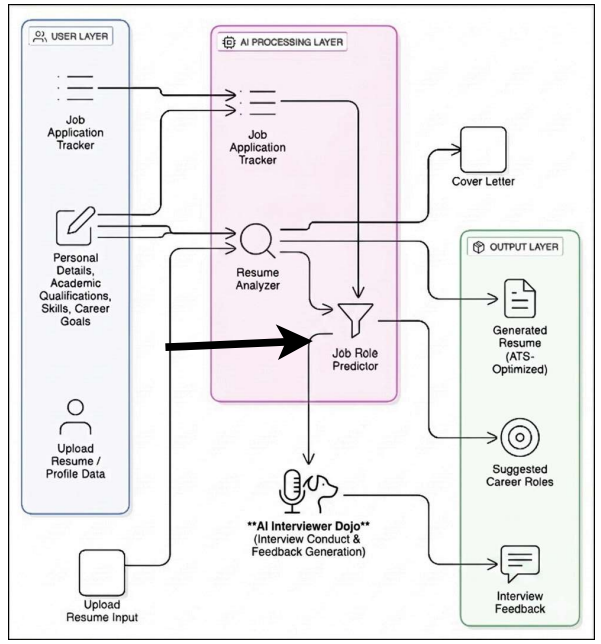
- Proposed solution: an AI-powered platform offering resume building, cover letter creation, and interview practice in one place.
- Modular system with AI/NLP support provides personalized guidance, improving confidence and job readiness.

OBJECTIVE

- To develop a user-friendly web application that helps students create professional career documents with ease.
- To generate personalized resumes and cover letters using AI-based suggestions tailored to a student's skills and background.
- To provide an AI-powered job-role recommendation feature that guides students toward suitable career options. .
- To build a Job Application Tracker that allows students to manage and monitor all their internship and job applications in one place.
- To offer all these features within a single integrated platform, making the placement preparation process more organized, efficient, and accessible.

METHODOLOGY

- **User Layer:** Users input personal details, skills, career goals, and upload resumes; also track job applications.
- **AI Processing Layer:** Core modules include Resume Analyzer, Job Role Predictor, and Job Application Tracker to process and evaluate data. tailored to a student's skills and background.
- **AI Features:** Generates ATS resumes, cover letters, and conducts mock interviews with feedback.
- **Output Layer:** Provides optimized resume, career role suggestions, and interview insights.



IMPLEMENTATION DETAILS

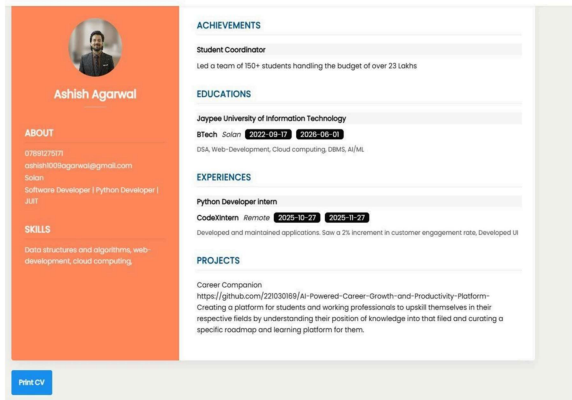
Tools and Technologies: The proposed system was created using.

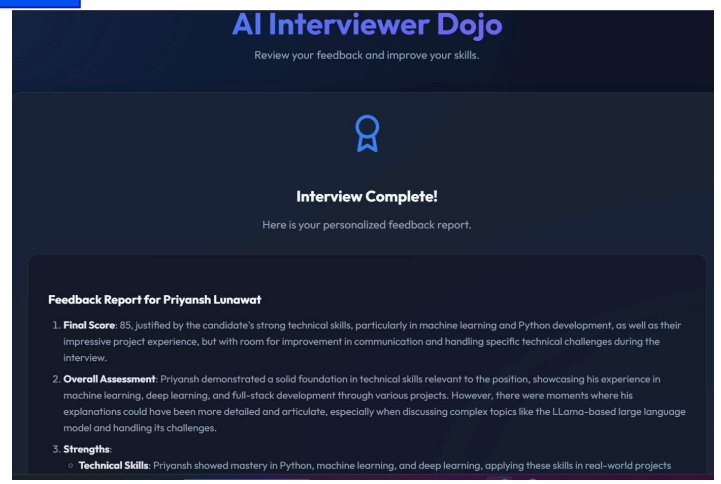
- **Frontend:** React.js / HTML / CSS
- **Backend**: Node.js or Django (depending on implementation),Python, Javascript
- **Database**: MongoDB / MySQL
- **Resume Generation:** Template-driven PDF generation and through Ollama Llama 3.x model
- **AI interview :** Groq API, speech recognition
- **API Integration:** RESTful API architecture
- **Authentication:** JWT / Session-based login
- **UI/UX Design:** Clean, modern, responsive interface

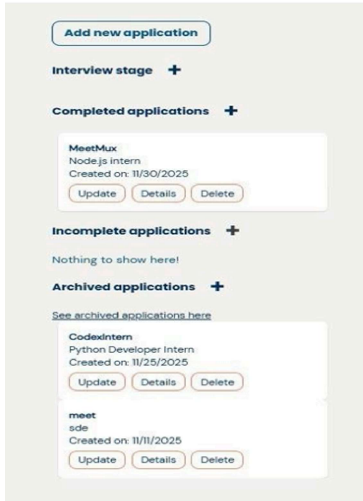
Workflow of Key Features:

- **Resume Generator + Job Prediction:** User fills form→ AI generates resume → Predicts future job roles .
- **Cover Letter Generator:** User selects job → AI drafts personalized cover letter.
- **Job Application Tracker :** allows users to add and manage their job applications by entering details like company, role, and application status. The system continuously updates progress and displays all applications on a unified dashboard withfilters.
- **AI Interviewer Dojo:** User inputs profile/resume → AI generates questions → conducts mock interview → provides feedback & improvement tips.

RESULTS







Models	Accuracy (%)	Precision	Recall	F1-Score
Rule-based parser	82.5	0.81	0.79	0.80
Random forest/SVM	89.6	0.88	0.87	0.87
CNN	90.8	0.90	0.89	0.89
ATS Commercial parser	93.4	0.94	0.93	0.93
Proposed Hybrid Model	95.8	0.96	0.95	0.95

PUBLICATION

P. Lunawat, N. Mittal, U. Sharma, A. Agarwal, and A. Sharma, "AI-Powered Career Growth and Productivity Platform," *International Journal for Research Trends and Innovation (IJRTI)*, communicated for publication, 2026. Status: Accepted.

CONCLUSION

- Unified AI platform integrates resume building, cover letters, interview prep, job recommendations, and application tracking.
- Enhances job readiness by providing personalized guidance, reducing stress, and streamlining the transition from academics to careers.

FUTURE SCOPE

- Unified AI platform integrates resume building, cover letters, interview prep, job recommendations, and application tracking.
- Enhances job readiness by providing personalized guidance, reducing stress, and streamlining the transition from academics to careers.

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